GameVault

User Manual and Setup Guide

Edition 14 · October 2026 · Covers NES and Famicom Disk System, SG-1000 and Master System, Apple II, Genesis / Mega Drive, Super Nintendo, PlayStation, Nintendo 64 and PlayStation 2

> GameVault never ships, downloads or links to commercial games, BIOS files or system ROMs. It plays only files you supply yourself, such as dumps of cartridges and discs you own, or freely licensed homebrew.

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Introduction

GameVault is a Windows app for playing and collecting classic console and computer games. It emulates the original hardware accurately, sets itself up with no configuration, and keeps a library of your collection: what you own, what you've beaten, your playtime, notes, passwords and save states.

GameVault plays games **you supply yourself**, such as dumps of cartridges and discs you own, or freely licensed homebrew. It never includes, downloads or links to commercial games, BIOS files or system ROMs. It does come with four freely licensed NES homebrew games, so there is something to play the first time you start it (see Bundled homebrew).

Supported systems at a glance

System	Game files	Needs your own BIOS or system ROM?
Nintendo Entertainment System (NES) / Famicom	`.nes`	No
Famicom Disk System	`.fds`	**Yes**: `disksys.rom` (8 KB)
Sega SG-1000 / SC-3000	`.sg`, `.sc`	No
Sega Master System	`.sms`	No
Apple II, II+, IIe	`.dsk`, `.do`, `.po`, `.nib`, `.woz`	**Yes**: system ROM (12 or 16 KB) and Disk II boot ROM (256 bytes)
Sega Genesis / Mega Drive	`.md`, `.gen`, `.smd`, `.bin`	No
Super Nintendo / Super Famicom	`.sfc`, `.smc`	Only for DSP-chip games: that chip's program ROM, such as `dsp1b.rom` (8 KB)
Sony PlayStation	`.cue` + `.bin`, `.chd`, `.m3u`, `.exe`	**Yes**: a 512 KB console BIOS such as SCPH-1001
Nintendo 64	`.z64`, `.v64`, `.n64`	No
Sony PlayStation 2	`.iso`, `.chd`, `.cue` + `.bin`, `.elf`	**Yes, for discs**: a 4 MB console BIOS such as SCPH-39001. Homebrew `.elf` programs run without it.

Cartridge games can also be inside a `.zip` file. Details for each system, including what isn't supported yet, are in Supported systems.

What you must supply

GameVault works out of the box, but some things only you can provide. This checklist covers everything.

Item	Required?	Where it goes	More
Your games (ROM files and disc images)	Yes, beyond the bundled homebrew	Any folder; add it in the Library	Quick start
Famicom Disk System BIOS (`disksys.rom`, 8 KB)	Only for `.fds` games	Settings → Disk System	Famicom Disk System
Apple II system ROM (12 KB for II/II+, 16 KB for IIe) and **Disk II boot ROM** (256 bytes)	Only for Apple II	Settings → Apple II	Apple II
Super Nintendo DSP chip ROMs (`dsp1b.rom` and the others, 8 KB each)	Only for games with a DSP chip, such as *Super Mario Kart*	Settings → Super Nintendo	Super Nintendo
PlayStation BIOS (512 KB, e.g. SCPH-1001)	Only for PlayStation discs	Settings → PlayStation	PlayStation
PlayStation 2 BIOS (4 MB, e.g. SCPH-39001)	Only for PlayStation 2 discs	Settings → PlayStation 2	PlayStation 2
Catalog DAT files (Logiqx XML, such as No-Intro or Redump)	Optional: names your games and gives a complete checklist	Library → Import catalog	Importing a catalog
Cover pictures	Optional	Game Details → Choose a picture…	Game Details
Game controller	Optional; the keyboard works	Plug it in	Controls
RetroAchievements account	Optional	Settings → RetroAchievements	RetroAchievements
Code collection (`.xlsx` or `.csv`)	Optional, NES codes	Settings → Codes	Code collections
INL Retro-Prog dumper, its software and USB driver	Optional, for dumping NES cartridges	Settings → Cartridge dumper	Cartridge dumper

Item	Required?	Where it goes	More
A second PC on the same network with the same game file	Optional, for netplay	Player → Netplay	Netplay

> **Make your own dumps.** BIOS files, system ROMs and games are copyrighted. Dump them from hardware you own. The NESdev, Apple II and PlayStation communities document how to do this for each system.

About this manual

This manual describes GameVault as of the edition on the title page. Keys are shown like `F5`, and screen labels in **bold**. Paths like `%LOCALAPPDATA%\GameVault\` can be pasted into the File Explorer address bar.

Installing GameVault

System requirements

	Minimum
Operating system	Windows 10 version 1809 or later, or Windows 11
Processor	x64 (recommended), ARM64 or 32-bit x86. A 4-core CPU from 2018 or later runs the 8- and 16-bit systems comfortably. PlayStation and Nintendo 64 need a faster CPU; the N64's graphics use extra cores when there are some. PlayStation 2 games need a fast CPU with 4 or more cores, and many still run below full speed.
Memory	4 GB. GameVault itself uses under 400 MB with the default rewind memory.
Graphics	Any GPU with Direct3D 11
Audio	Any Windows audio device
Disk	About 300 MB for the app, plus your games
Optional	A USB or Bluetooth game controller; an internet connection for RetroAchievements and GameGenie.com codes; an INL Retro-Prog for dumping cartridges

GameVault is a portable (unpackaged) app. It doesn't need administrator rights, doesn't use the Microsoft Store, and keeps all its data in `%LOCALAPPDATA%\GameVault\`.

Option A: a ready-made copy

If you were given a GameVault folder or `.zip`:

1. Extract it anywhere you like, for example `C:\Games\GameVault`.
2. Run `GameVault.exe`.
3. If Windows SmartScreen says *Windows protected your PC*, choose **More info → Run anyway**. This appears for apps that aren't code-signed.

To uninstall, delete the folder. To also remove your library, saves and settings, delete `%LOCALAPPDATA%\GameVault\` (back it up first if you want to keep your saves).

Option B: build it from source

You need:

Tool	Where to get it	Needed for
.NET 10 SDK	https://dotnet.microsoft.com/download or `winget install Microsoft.DotNet.SDK.10`	Always
Git	`winget install Git.Git`	Getting the source
CMake and a C compiler	Visual Studio 2022 or later with the *Desktop development with C++* workload, or `winget install Kitware.CMake` plus the Visual Studio Build Tools	RetroAchievements only. Without them GameVault still builds and runs, with RetroAchievements turned off.
Visual Studio with the *Windows application development* workload	https://visualstudio.microsoft.com	Optional, for editing the app

Then, in PowerShell:

```
git clone https://github.com/ChrisCorea26/GameVault.git
cd GameVault
dotnet build GameVault.sln -c Release
dotnet run --project src/GameVault.App -c Release
```

To make a stand-alone folder you can copy to another PC (it doesn't need .NET installed):

```
dotnet publish src/GameVault.App -c Release -r win-x64 --self-contained -o publish\GameVault
```

Then run `publish\GameVault\GameVault.exe`. Use `-r win-arm64` or `-r win-x86` for other processors.

> **Note:** The first build compiles the RetroAchievements component with CMake. If CMake or a C compiler is missing, you'll see a build warning and RetroAchievements will say *The RetroAchievements component is missing from this copy of GameVault*. Everything else works.

Trying GameVault without any games

To look around before you have any games, start GameVault with sample data:

```
GameVault.exe --sample-data
```

or, from the source folder, `dotnet run --project src/GameVault.App -- --sample-data`. This fills every screen with a made-up library, save states and passwords, and shows a *Sample data* badge in the title bar. Nothing is saved. `--sample-data=empty` shows the first-run screens with an empty library.

Firewall prompt

The first time you host or join a netplay game, Windows Defender Firewall may ask whether GameVault can use the network. Allow it on **Private networks** for netplay to work. See Netplay.

Quick start

You can be playing within a minute, without opening Settings.

1. **Start GameVault.** The Library opens with four homebrew games that come with GameVault (see Bundled homebrew). You can play one of those straight away.
2. **Choose Add a ROM folder** and pick the folder where you keep your games. GameVault searches it and every folder inside it, and lists every game it finds.
3. **Click a game**, then **Play**. Or right-click its tile and choose **Play**.
4. **Play** with the keyboard (arrow keys, `Z` = B, `X` = A, `Enter` = Start, `Right Shift` = Select) or plug in a controller. It's set up automatically.
5. **Press** `Esc` to pause. **Back to library** saves where you are, and next time GameVault offers to resume.

To play a single file without adding its folder, choose **Open game file**.

Next steps

- **Name your games and see a complete checklist** by importing a catalog. See Importing a catalog.
- **Set up BIOS files** if you play Famicom Disk System, Apple II, PlayStation or PlayStation 2 games. See Supported systems.
- **Learn the hotkeys**: `F5` save state, `F8` load it, hold `Backspace` to rewind, hold `Tab` to fast-forward, `F11` full screen. See Hotkeys.
- **Mark what you own** and track your progress in Game Details.

The screens

Screen	How to get there	What it's for
Library	Navigation pane, or `Alt`+`L`	Your games and the complete checklist
Game Details	Click a game	Collection status, notes, save states, passwords, journal, codes, per-game properties
Player	**Play**	The game itself, with save states, assist, practice, netplay and more
Settings	The gear in the navigation pane	Folders, video, audio, controllers, BIOS files, family profiles, RetroAchievements

Screen	How to get there	What it’s for
Import a catalog	Library → **Import catalog**	Adds a list of known releases
Dump a cartridge	Library → **Dump cartridge**	Dumps an NES cartridge with an INL Retro-Prog

Supported systems

Each section below follows the same pattern: the files GameVault reads, what you must supply, setup, controls, where saves go, and what isn't supported yet.

Things that apply to every system:

- GameVault detects the system from the file, so you never pick it yourself.
- Cartridge games can be inside a `.zip` that holds exactly one game. Files larger than 64 MB are ignored. PlayStation discs can't be zipped.
- For BIOS and ROM settings, choose **Choose...** next to the box and pick the file, or paste its full path (for example, in File Explorer Shift+right-click the file and choose **Copy as path**).
  - A BIOS you pick with **Choose...** (Disk System, PlayStation, PlayStation 2) is checked and copied into GameVault's own `bios` folder (see Your data), and the box then shows the copy. So the BIOS keeps working if you move the original or unplug the drive it's on. A file that fails the check isn't copied.
  - A pasted path, and the Apple II ROMs, are read from where they are each time.
  - Files are checked by size only (the PlayStation 2 BIOS by its version too).
- Battery saves, memory cards and disk writes go to `%LOCALAPPDATA%\GameVault\Saves\`. They're matched to the game by its contents, so renaming or moving a game file keeps its saves. With **Settings → Gameplay → Keep game saves next to the game files**, cartridge saves go beside each game as a `.sav` instead.
- Save states, rewind, practice, challenges and netplay work on every system unless noted.

Nintendo Entertainment System (NES)

Files	`.nes` (iNES and NES 2.0 headers)
You supply	Nothing but your games
Saves	Battery saves as `.sav`

Controls: the standard layout in Controls, with turbo A and B. Zapper light-gun games and the Arkanoid paddle use the mouse, and the Power Pad mat uses the keyboard (see Power Pad). The Four Score adds players 3 and 4: the third and fourth controllers you connect join as Players 3 and 4 while the game runs, and **Use for Player 3 / 4** moves one there. GameVault recognises about 65 Zapper, Four Score, Power Pad and Arkanoid releases and sets their controls itself; for others choose them in **Game Details → Properties → Controls**.

Game data GameVault ships: memory maps for *Super Mario Bros.*, *The Legend of Zelda* (USA, first release) and *Metroid* (USA), which give F2's "before you lost a life" (Super Mario Bros.), the explored map (Zelda), the stat overlay and challenges; and automatic password capture for *Metroid*

(USA). They were written from the games' published disassemblies and are matched by the game's exact dump, so other releases (another region or revision) aren't covered.

NES-only features: Game Genie and RAM codes, password capture, the Zapper, and cartridge dumping.

Compatibility: GameVault supports the cartridge boards (mappers) that cover the large majority of the licensed library, including NROM, MMC1, UxROM, CNROM, MMC3, AxROM, MMC2, MMC4, MMC5, VRC2/4/6/7, FME-7, Namco 163 and more, with their expansion sound. A game on an unsupported board shows *This game uses a cartridge type GameVault doesn't support yet (mapper N)*.

If a game won't start: *That file isn't an NES game, or it's damaged* usually means a bad dump or a file with the wrong extension.

Famicom Disk System

Files	`.fds`, with or without the 16-byte fwNES header
You supply	**The Disk System BIOS**, `disksys.rom`, exactly **8 KB**, dumped from your own Disk System
Saves	Disk writes as `.fdssave` patches; your disk image is never changed

Setup:

1. Put `disksys.rom` anywhere on your PC, for example `C:\NES\disksys.rom`.
2. Open **Settings → Disk System**, choose **Choose...** and pick the file (or paste its path into the box).
3. Open an `.fds` game.

Changing disk sides: when the game asks you to turn the disk over, press `F4` for the next side, or pick one from the disk drop-down in the Player (*No disk*, *Disk 1, side A*, *Disk 1, side B*, ...).

Not supported yet: netplay with Disk System games.

Messages:

Message	What to do
To play it, choose your own Disk System BIOS (disksys.rom) in Settings → Disk System.	No BIOS is set. Follow the setup above.
GameVault couldn't open your Disk System BIOS.	The file moved or was renamed. Check the path.

Message	What to do
The Disk System BIOS (disksys.rom) must be exactly 8 KB.	The file isn't the BIOS, or the dump is bad.

Sega SG-1000 and Master System

Files	`.sms` (Master System); `.sg`, `.sc` (SG-1000 / SC-3000)
You supply	Nothing but your games. No BIOS is needed.
Saves	Battery saves as `.sav`

Controls: the NES layout. **B** (Z) is **Button 1**, **A** (X) is **Button 2**, and **Start** (Enter) is the console's **Pause** button. Select does nothing. **Remap…** shows these buttons by their Sega names.

Region: games run at NTSC speed (60 Hz). For a European game, set **Game Details → Properties → Region** to *PAL (50 Hz)*.

Not supported yet: the SC-3000 keyboard (so BASIC cartridges can't be used) and FM sound for Japanese Master System games.

Apple II, II+ and IIe

Files	5.25-inch disk images: `.dsk`, `.do`, `.po` (143,360 bytes), `.nib` (232,960 bytes), `.woz` (WOZ 1 and 2)
You supply	**The system ROM** from your Apple (12 KB for a II or II+, 16 KB for a IIe) **and the Disk II controller's boot ROM** (256 bytes, the P5 PROM 341-0027)
Saves	Disk writes as `.a2save` patches; your disk images are never changed

Setup:

1. Put both ROM files anywhere, for example `C:\Apple\apple2e-enhanced.rom` and `C:\Apple\disk2.rom`.
2. Open **Settings → Apple II** and choose each file with **Choose…** next to **System ROM** and **Disk II boot ROM** (or paste their paths).
3. Open a disk image.

Which Apple runs: a 16 KB ROM runs an enhanced Apple IIe (65C02 processor); a 12 KB ROM runs an Apple II+. For a game that needs another model (for example an unenhanced IIe), set **Game Details → Properties → Apple II model**.

> **Important:** Without the Disk II boot ROM the computer starts with no disk controller, so disks can't boot. GameVault then shows *This disk can't start without the Disk II boot ROM. Choose your own Disk II boot ROM (256 bytes) in Settings → Apple II. GameVault doesn't include it.*

Keyboard: while an Apple II game runs, the whole keyboard types into the computer.

Apple key	PC key
Esc	F1
Left arrow	Backspace
Reset	Ctrl + F12
Rewind (GameVault)	Shift + Backspace

Disks: the Player's **Disks** menu shows drives 1 and 2, with **Insert disk in drive 1...** / **Eject drive 1** and the same for drive 2. Use it for games on more than one disk.

Netplay: both players need the same ROM files as well as the same disk.

Not supported yet: loading an older save state doesn't undo later disk writes; no monochrome display option; your own character ROM isn't used; video mode changes in the middle of a frame aren't shown.

Messages:

Message	What to do
To play it, choose your own Apple II ROM files in Settings → Apple II.	Set the System ROM path.
The Apple II ROM must be 12 KB (II/II+) or 16 KB (IIe).	Wrong file or bad dump.
The Disk II boot ROM must be 256 bytes.	Wrong file.
This disk can't start without the Disk II boot ROM. Choose your own Disk II boot ROM (256 bytes) in Settings → Apple II. GameVault doesn't include it.	Set the Disk II boot ROM path.
A .dsk, .do or .po disk image is 143,360 bytes.	The image is damaged or isn't a 5.25-inch disk.
GameVault reads 5.25-inch WOZ images; this one is for 3.5-inch disks.	3.5-inch disks aren't supported.

Sega Genesis / Mega Drive

Files	`.md`, `.gen`, `.smd`, `.bin` (a `.bin` is used only if it has a Sega header)

You supply	Nothing but your games. No BIOS is needed.
Saves	Battery saves (SRAM) as `.sav`

Region is read from the game automatically. To run a game as another region (for example a European game at 50 Hz), set **Game Details → Properties → Region**.

Controls: GameVault uses the 6-button pad. For the few older games that misread it, set **Game Details → Properties → Controller** to *3-button*.

Genesis	Keyboard	Controller (position)	Xbox label
A, B, C	Z, X, C	Left, bottom, right	X, A, B
X, Y, Z	Q, W, E	Left shoulder, top, right shoulder	LB, Y, RB
Mode	Right Shift	Back	View
Start	Enter	Start	Menu

While a Genesis game runs the shoulder buttons are game buttons, so use Backspace and Tab to rewind and fast-forward.

Not supported yet: interlaced high-resolution mode, Master System mode, EEPROM saves, Sega CD and 32X.

Super Nintendo

Files	`.sfc`, `.smc` (a 512-byte copier header is skipped)
You supply	Your games. Games with a **DSP chip** also need that chip's program ROM, dumped from your own cartridge (see below). GameVault uses its own boot program, so no other Nintendo ROM is needed.
Saves	Battery saves as `.sav` (Super FX games' saves too)

Controls:

SNES	Keyboard	Controller (position)	Xbox label
B, A	Z, X	Bottom, right	A, B
Y, X	A, S	Left, top	X, Y
L, R	1, 2	Left shoulder, right shoulder	LB, RB
Select, Start	Right Shift, Enter	Back, Start	View, Menu

There's no turbo on the SNES. Use `Backspace` and `Tab` to rewind and fast-forward.

Enhancement chips. Some cartridges carry an extra chip:

- **Super FX** (*Star Fox*, *Yoshi's Island*, *Stunt Race FX*, …): works with nothing extra. The game's own code runs on it.
- **DSP-1** (*Super Mario Kart*, *Pilotwings*, …), DSP-2, DSP-3 and DSP-4: the chip runs a program built into it, which GameVault doesn't include. Dump it from your own cartridge and name the file `dsp1b.rom` (or `dsp1.rom`), `dsp2.rom`, `dsp3.rom` or `dsp4.rom`. Put it in a folder, then choose that folder in **Settings → Super Nintendo → Enhancement chip ROMs**. Either byte order works. DSP-2, DSP-3 and DSP-4 haven't been tried with real games yet.

Message	What to do
This game needs the DSP-1 chip's program ROM. Put your own dump (dsp1b.rom or dsp1.rom) in the folder chosen in Settings → Super Nintendo → Enhancement chip ROMs. GameVault doesn't include it.	Add the file and choose its folder, then open the game again.
This game needs the … chip, which GameVault doesn't support yet.	The chip isn't emulated yet.

Not supported yet: the SA-1, S-DD1, S-RTC, OBC1, Cx4, SPC7110 and ST01x chips, the Super Game Boy and the Satellaview; interlaced modes; effects that change the picture partway along a line.

Sony PlayStation

Files	`.cue` with its `.bin` tracks; `.chd` (version 5); `.m3u` playlists for multi-disc games; `.exe` (PS-EXE homebrew)
You supply	**A PlayStation BIOS**, 512 KB, dumped from your own console. SCPH-1001 (North America), SCPH-5500 (Japan) and SCPH-7502 (Europe) are typical.
Saves	One memory card per game in slot 1, as `.mcd`

Setup:

1. Put the BIOS file anywhere, for example `C:\PlayStation\scph1001.bin`.
2. Open **Settings → PlayStation**, choose **Choose…** and pick the file (or paste its path into the box).
3. Optionally turn on **Skip the startup screens** to start games straight away. A few games need the intro to start correctly; turn it off if one doesn't.
4. Open the game's `.cue`, `.chd` or `.m3u` file.

> **Note:** GameVault reports each disc's real region to the BIOS, as a real console does, so use a BIOS from the same region as your discs.

Disc images:

- **`.cue` + `.bin`**: keep the `.cue` and all its `.bin` files together in one folder. Open the `.cue`. A `.bin` without a `.cue` is ignored. Audio tracks must be raw `.bin`, not `.wav` or `.mp3`.
- **`.chd`**: must be CHD version 5. Convert older files with MAME's tool: `chdman copy -i old.chd -o new.chd`. CHDs that depend on a parent CHD must be merged the same way.
- **Multi-disc games**: make an `.m3u` text file next to the discs with one disc file per line, for example:

  ```
  Final Fantasy VII (Disc 1).cue
  Final Fantasy VII (Disc 2).cue
  Final Fantasy VII (Disc 3).cue
  ```

  Open the `.m3u`. When the game asks for the next disc, press `F4` or pick it from the disc drop-down (*No disc (shell open)*, *Disc 1: …*, *Disc 2: …*). Lines starting with `#` are ignored.
- **`.exe` homebrew** runs without a BIOS unless it calls into the BIOS.

Discs are read from where they are; they're never copied or loaded into memory whole.

Controls (DualShock):

PlayStation	Keyboard	Controller (position)	Xbox label
Cross, Circle	Z, X	Bottom, right	A, B
Square, Triangle	A, S	Left, top	X, Y
L1, R1	1, 2	Left shoulder, right shoulder	LB, RB
L2, R2	3, 4	Left trigger, right trigger	LT, RT
L3, R3 (analog mode)	5, 6	Click the left stick, click the right stick	LS, RS
Select, Start	Right Shift, Enter	Back, Start	View, Menu
D-pad	T F G H	D-pad	D-pad
Left stick	Arrow keys (full push)	Left stick	Left stick
Right stick	I J K L (full push)	Right stick	Right stick

Analog mode: the controllers are DualShocks. They start digital, and the left stick (and the arrow keys) work the D-pad. Games that use the sticks switch them to analog themselves. For a game that supports the sticks but waits for you to press the DualShock's Analog button, turn on **Settings → PlayStation → Analog mode**. Games that rumble shake your controller, if it has motors.

Light gun and mouse: for a GunCon game (*Time Crisis*, *Point Blank*), set **Game Details → Properties → Port 1** to **GunCon (the mouse aims)**: point with the mouse, the left button is the trigger, the right button A and the middle button B. Most GunCon games start with a calibration screen; shoot its centre. For a mouse game (*Lemmings*, *Discworld*), choose **Mouse**: moving the mouse over the picture moves it, with its left and right buttons.

Memory cards: each game gets its own card in slot 1, so you never run out of space. Slot 2 is empty.

Videos and other pictures in 24-bit colour are shown with all their colours.

Not supported yet: multitap, the Justifier light gun, link cable, GameShark codes, a memory card manager, and enhanced (upscaled) graphics.

Messages:

Message	What to do
To play it, choose your own PlayStation BIOS in Settings → PlayStation.	Follow the setup above.
…isn't a 512 KB BIOS file.	Wrong file or bad dump.
The cue sheet '…' names '…', which isn't in its folder.	A `.bin` was renamed or moved. Keep the names in the `.cue` and the files the same.
…is a version N CHD. GameVault reads version 5…	Convert it with `chdman copy`.
This CHD needs its parent CHD…	Merge it with `chdman copy`.

Nintendo 64

Files	`.z64`, `.v64`, `.n64` (any byte order), up to 64 MB
You supply	Nothing but your games. No BIOS or PIF ROM is needed.
Saves	Cartridge saves (EEPROM, SRAM, FlashRAM) as `.sav`; one Controller Pak per game as `.mpk`

Settings → Nintendo 64 → Expansion Pak (on by default) gives games the extra 4 MB of memory. Turn it off for the rare game that won't start with it.

Save type: GameVault picks each game's save type from its database (`data\gamedb\n64\saves.json` in the GameVault folder) and from homebrew headers. Games it doesn't know get a 4 Kbit EEPROM. If a game doesn't keep its save, its save type is probably missing from the database: choose the cartridge's save type in **Game Details → Properties → Save type**.

Controls:

N64	Keyboard	Controller
Analog stick	Arrow keys (full push)	Left stick
A, B	Z, A	Bottom, left (Xbox A, X)
C buttons	I J K L (also X = C-down, S = C-left)	Right stick; right and top buttons are C-down and C-left
Z	3 or 4	Either trigger
L, R	1, 2	Left shoulder, right shoulder
Start	Enter	Start
D-pad	T F G H	D-pad

Performance: the N64's graphics are drawn in software, on several processor cores at once, so a CPU with more cores helps. 2D games and most 3D scenes run at full speed on a modern CPU; the heaviest 3D scenes may still slow down on a CPU with few cores.

Rumble Pak: turn on **Settings → Nintendo 64 → Rumble Pak** and every controller gets one, so games that use it shake your controller (if it has rumble motors). Games can't save to the Controller Pak while it's on.

Games that draw in 32-bit colour are shown with all their colours, without the banding of 16-bit colour.

Not supported yet: Transfer Pak, mouse, voice recognition unit, 64DD, GameShark codes, and the N64's video filters.

Sony PlayStation 2

Files	`.iso` (DVD or CD images); `.chd` (version 5, CD or DVD); `.cue` with its `.bin` tracks (CD games); `.elf` (homebrew programs)
You supply	**A PlayStation 2 BIOS**, 4 MB, dumped from your own console (for example SCPH-39001), for discs. Homebrew `.elf` programs run without one.

Saves	One 8 MB memory card per game in slot 1, as `.ps2`

Setup:

1. Put the BIOS file anywhere, for example `C:\PlayStation 2\scph39001.bin`.
2. Open **Settings → PlayStation 2**, choose **Choose...** and pick the file (or paste its path into the box). GameVault checks the file and shows its version, for example *PlayStation 2 BIOS version 1.60, Europe, 2002-02-07.*
3. Optionally turn on **Skip the startup screens** to start discs straight away instead of showing the console's startup screens and browser. A few games need them; turn it off if one doesn't start.
4. Open the game's `.iso`, `.chd` or `.cue` file, or an `.elf` program.

GameVault tells PlayStation and PlayStation 2 discs apart by what's on them, so a `.cue`, `.chd` or `.iso` can be either; it shows up under the right system in the library.

Homebrew `.elf` programs run on GameVault's own built-in system software when no BIOS is set. With a BIOS set they start through it instead, which programs that load their own IOP modules need.

Disc images: as for the PlayStation, keep a `.cue` with its `.bin` files, and use version 5 CHDs. Discs are read from where they are; they're never copied or loaded into memory whole.

Controls (DualShock 2):

PlayStation 2	Keyboard	Controller (position)	Xbox label
Cross, Circle	Z, X	Bottom, right	A, B
Square, Triangle	A, S	Left, top	X, Y
L1, R1	1, 2	Left shoulder, right shoulder	LB, RB
L2, R2	3, 4	Left trigger, right trigger	LT, RT
L3, R3	5, 6	Click the left stick, click the right stick	LS, RS
Select, Start	Right Shift, Enter	Back, Start	View, Menu
D-pad	T F G H	D-pad	D-pad
Left stick	Arrow keys (full push)	Left stick	Left stick
Right stick	I J K L (full push)	Right stick	Right stick

The pressure-sensitive buttons report full presses. While a PlayStation 2 game runs the left shoulder button is L1, so use Backspace to rewind.

Memory cards: each game gets its own 8 MB card in slot 1, created already formatted and saved with your other game saves. Slot 2 is empty.

Rewind, run-ahead and netplay: a PlayStation 2 is much larger than the other consoles, so:

- **Rewind** goes back in half-second steps.
- **Reduce input lag** (run-ahead) is always off for PlayStation 2 games.
- **Netplay** isn't available for PlayStation 2 games.

Performance: the PlayStation 2 is emulated in software. GameVault translates the main processor's code into your PC's own as it runs, and draws the graphics on several processor cores at once, so a CPU with more cores helps. Lighter games can reach full speed on a fast PC, but many games still run below it, especially 3D games that lean on the PS2's vector units. Enhanced graphics and more speed-ups are planned.

Not supported yet: changing discs in multi-disc games (an `.m3u` starts its first disc), vibration, the multitap, PlayStation games on the PlayStation 2, USB devices, the network adapter and hard drive, GameShark codes, memory card slot 2, and enhanced (upscaled) graphics.

Messages:

Message	What to do
This is a PlayStation 2 disc. To play it, choose your own PlayStation 2 BIOS in Settings → PlayStation 2. GameVault doesn't include it.	Follow the setup above.
The PlayStation 2 BIOS chosen in Settings → PlayStation 2 isn't a 4 MB BIOS file. Choose the BIOS dumped from your own console.	Wrong file or bad dump.
GameVault couldn't open your PlayStation 2 BIOS. Check the file chosen in Settings → PlayStation 2.	The file was moved, deleted or is locked.
That file isn't a PlayStation 2 program (ELF) or disc, or it's damaged.	Check the file; re-dump the disc if needed.
Netplay isn't available for PlayStation 2 games: its rollback saves and loads the whole console every frame, and a PlayStation 2 is too large for that.	Play it locally.

The Library

The Library is GameVault's home screen. It shows the games in your folders, and, once you've imported a catalog, a complete checklist of every known release with the ones you own highlighted.

Adding games

Add a ROM folder (Alt + A) opens a folder picker. GameVault searches the folder and all its subfolders for every supported file type (see Supported systems), hashes each game, and matches it against your catalog. The status line shows *Looking for games…*, then *Checking N of M files…*, then *Found N games*.

- Add as many folders as you like. Manage the list in **Settings → Library**, where **Look for new games** rescans every folder.
- While GameVault is open it watches your folders, so files you add or delete appear and disappear automatically.
- A `.zip` file is read if it contains exactly one game (not a PlayStation disc).
- Games that aren't in your catalog are listed as **Unknown / Homebrew**. You can give them a title, publisher, developer, year, genre, region and number of players in Game Details, so they can be filtered and grouped like any other game.

Open game file (Alt + O) plays a single file straight away, without adding its folder.

Bundled homebrew

GameVault comes with four freely licensed NES homebrew games by Damian Yerrick. On first launch it adds their folder (`data\homebrew` in the GameVault folder) to your library once, with their names and details filled in.

Game	What it is	Players	License
Thwaite	Defend the villages from incoming missiles	1–2	GPL 3 or later (its source is included)
Concentration Room	The memory card game	1–2, or against the computer	GPL 3 or later
RHDE: Furniture Fight	Build and furnish your house, then defend it	1–2	GNU All-Permissive
robotfindskitten	Look through the room's odd objects until you find kitten	1	zlib

Each game's folder has its `LICENSE` and a `README.md` saying where it came from. If you don't want them, remove the folder in **Settings → Library**; GameVault doesn't add it again.

Browsing

Control	What it does
Search (Alt + S)	Matches title, publisher or genre.
All games / Owned only	Show only games you've marked as owned.
My games / Complete checklist	*My games* shows games you have a file for or own. *Complete checklist* shows every game in your catalog.
System	Appears when your library has more than one system.
Region, Genre, Publisher, Year, Status	Filters. Each starts at *Any*.
List view / grid (Alt + V)	Switches between cover tiles and a compact list that also shows playtime.

The bar at the top shows your collection progress, for example *214 / 677 licensed releases owned*. Only licensed releases in your catalog are counted, so this appears after you import a catalog.

Click a game to open its details. Right-click a tile and choose **Play** to start it directly.

Importing a catalog

A catalog is a list of every known release of a system. It lets GameVault name your games correctly and show a complete checklist. GameVault does **not** ship one: you download it yourself (see What you must supply).

1. Choose **Import catalog** (Alt + I).
2. Choose **Choose a DAT file…** and pick a `.dat` or `.xml` file in the Logiqx XML format, such as a No-Intro DAT for the NES, SG-1000, Master System, Apple II, Genesis, Super Nintendo or Nintendo 64, or a Redump DAT for the PlayStation or PlayStation 2. A file that lists no games GameVault plays shows *That file doesn't list any games GameVault plays.*
3. Check the preview, for example *"Nintendo - NES" lists 3,412 games (1,980 licensed releases).*, and choose **Import**.
4. Games in your folders that match are now named and grouped.

Import one catalog per system. GameVault recognises the system from the DAT's name (for example *Sega - Mega Drive - Genesis* or *Sony - PlayStation*); DATs it doesn't recognise are treated as NES. Importing a newer version of the same DAT replaces the old entries.

System	Where to get a catalog
NES, Master System, SG-1000, Genesis, SNES, N64, Apple II	No-Intro DATs (Logiqx XML), from the No-Intro DAT-o-MATIC site
PlayStation	Redump DATs (Logiqx XML), from the Redump site. Discs are matched on their first track.
PlayStation 2	Redump DATs (*Sony - PlayStation 2*). DVDs are matched on the whole image, CDs on their first track.

The older text-based clrmamepro DAT format isn't supported; choose the XML download.

> **Note:** A catalog holds names, hashes and release details only. It never contains games.

Game Details

Game Details is the page for a single game.

Cover, title and Play

- **Choose a picture...** sets your own cover (`.png` or `.jpg`), for example a photo of your cartridge. GameVault never downloads box art. **Remove** clears it.
- Games that aren't in the catalog let you edit the title, and **Edit details** opens boxes for **Publisher**, **Developer**, **Year**, **Genre**, **Region** and **Players**. They save when you leave each box.
- The details line shows region, number of players, whether the game is a licensed release, and the developer.
- Playtime and the last-played date are tracked automatically.
- **Play** (Alt + P) starts the game. It's disabled when GameVault has no file for the game, for example a checklist entry you own only on cartridge.

My collection

Every change saves immediately.

- **Cartridge**, **Box**, **Manual**, **Digital copy**: what you own, as separate check boxes.
- **Status**: Unplayed, Playing, Beaten or 100%.
- **Favorite** and **My rating** (1 to 5 stars).
- **Notes**: free text, for tips or where you bought the cartridge.

Save states

Your save states for this game, newest last. Click one to start the game from that point. Right-click a state, or use its **…** button, to **Load**, **Pin**, rename (**Name**, then **Save name**) or **Delete** it. Deleting shows an **Undo** button. See Save states and the timeline.

Properties

Per-game overrides. *Automatic* uses what GameVault's game database knows about the game. Changes apply the next time the game starts. Only the properties that apply to the game's system are shown.

Property	Systems	Values
Controls	NES	Automatic, Controller, Zapper, Four Score, Power Pad (keyboard), Arkanoid paddle (mouse)
Port 1	PlayStation	Automatic, DualShock, GunCon (the mouse aims), Mouse
Reduce input lag	All	Automatic, On, Off
Reduce flicker (more sprites per line)	NES	Automatic, On, Off
Region (speed)	NES	Automatic, NTSC (60 Hz), PAL (50 Hz), Dendy (50 Hz)
	Genesis	Automatic, USA (60 Hz), Japan (60 Hz), Europe (50 Hz)
	Super Nintendo	Automatic, NTSC (60 Hz), PAL (50 Hz)
	SG-1000, Master System	Automatic (NTSC, 60 Hz), NTSC (60 Hz), PAL (50 Hz)
Apple II model	Apple II	Automatic, Apple II+, Apple IIe, Enhanced Apple IIe
Controller	Genesis	Automatic (6-button), 6-button, 3-button (a few older games misread the 6-button pad)
Save type	Nintendo 64	Automatic, EEPROM (4 Kbit), EEPROM (16 Kbit), SRAM, Flash, None

Property	Systems	Values
Header corrections	NES	**Mapper** (a number, or empty to use the file's), **Mirroring** (Automatic, Horizontal, Vertical, Four-screen), **Battery-backed save** (Automatic, On, Off)

Use **Region** when a European game runs too fast or a US game too slow, **Save type** when an N64 game can't save, and **Header corrections** only when an NES file shows garbage or won't start because its header is wrong. A mapper that isn't a number from 0 to 4095 shows *The mapper is a number from 0 to 4095, or leave it empty to use the file's.*

GameVault's database sets **Controls** automatically for the NES's Zapper, Four Score, Power Pad and Arkanoid games (about 65 releases), so for those you usually don't need to change anything.

Passwords, Journal and Codes

- **Passwords**: passwords you captured with F9 or that GameVault saved automatically, with **Copy** and **Delete**. See Passwords.
- **Journal**: notes, screenshots (F12) and maps. See Journal and maps.
- **Codes**: Game Genie and RAM codes, shown for NES games only. See Game Genie and RAM codes.

Deleting a password, a code, a note or a picture asks first (*Delete this password?* and so on); choose **Delete** to go ahead or **Cancel** to keep it.

Dumping a cartridge

If you own an **INL Retro-Prog** USB cartridge dumper, GameVault can dump NES cartridges straight into your library. The dumper is optional. See Cartridge dumper for setup.

1. Choose **Dump cartridge** (Alt + D).
2. Search the catalog for the game (**Which game is it?**), or type its name if it isn't listed.
3. Pick the **Board**. **Program size (KB)** and **Graphics size (KB)** are filled in for you; change them only if you know your cartridge differs.
4. Choose **Next**, insert the cartridge label-facing-you, and choose **Dump**.
5. GameVault reads the cartridge up to 3 times to check the reads match, then checks the dump against the catalog. A good dump is saved as `<Title>.nes` in your first library folder (or `Documents\GameVault Dumps`), added to your library, and marked as owned (cartridge and digital copy).

If a read fails, clean the cartridge contacts and try again.

Playing games

The Player screen

The Player shows the game with a toolbar above it. Press Alt to see each button's access key.

Button	Key	What it does
Back to library	Alt + B	Saves a resume point and closes the game.
Pause / Resume	Alt + P, Esc	Pauses the game.
Reset	Alt + R	Presses the console's reset button.
Save states	Alt + S	The save-state timeline.
Assist	Alt + A	Slow motion, color-vision palettes and more.
Practice		Practice points and a segment timer.
Disk side / disc	F4	Famicom Disk System sides and multi-disc PlayStation games.
Disks	Alt + D	Apple II drives 1 and 2.
Codes	Alt + C	Game Genie and RAM codes (NES).
Netplay	Alt + N	Two players on the same network.
Controllers	Alt + O	The controller cards from Settings, so you can swap, reassign and remap without leaving the game. See Controllers in the Player.
Full screen	F11, Alt + Enter	Esc leaves full screen.
Stats	Alt + T	Shows values from the game, such as exact health. Only for games with a memory map.
Challenges	Alt + H	Make and take verified challenges. Only for games with a memory map.

Button	Key	What it does
Achievements	Alt + E	RetroAchievements progress, for recognised games.

Buttons that don't apply to the running game are hidden. The toolbar is hidden in full screen.

When you come back to a game, a bar offers **Resume** to pick up where you stopped. Close the bar to start fresh. After a break of a day or more, a **Previously on…** card shows how long you last played, your newest screenshot, map, note and password. Choose **Resume where I left off** or **Continue**.

If GameVault loses focus (for example you Alt + Tab away), the game pauses and resumes when you come back. You can turn this off in **Settings → Gameplay**.

Save states and the timeline

A save state captures the whole console at one moment. GameVault keeps as many as you like per game, each with a thumbnail, the date, and your playtime.

- F5 saves a state. F8 loads the newest one.
- **Save states → Save now** does the same as F5.
- Click a thumbnail to load it. Right-click to **Pin**, rename or **Delete**. Deleting can be undone.
- When you leave a game, GameVault saves a **Where you left off** state automatically.

Save states are separate from the game's own battery save (the in-game "save" of games like *The Legend of Zelda*). Battery saves are written automatically and kept in GameVault's data folder, not in your ROM folders, unless you turn on **Settings → Gameplay → Keep game saves next to the game files**.

> **Note:** A save state stores which codes were on. Save states can't be loaded in RetroAchievements hardcore mode or during a challenge attempt.

Rewind and fast-forward

- Hold Backspace or the controller's **left shoulder** button to rewind. A scrub bar appears; it spans the whole rewind history, however long. Release to carry on playing from there.
- To jump to a point, grab the scrub bar with the mouse while rewinding and drag it, back or forward. The game holds still while you drag, even if you let go of the rewind key. Let go of the bar to play on from that point.
- Hold Tab or the **right shoulder** button to fast-forward.
- F2 goes back to about 5 seconds before you last lost a life, for games with a memory map.

- How far back you can rewind depends on **Settings → Gameplay → Rewind memory** (256 MB by default, at least a minute of play).
- On the Apple II every key types into the computer, so use Shift + Backspace to rewind.

Controllers in the Player

Controllers opens the same cards as **Settings → Controllers**: one per player with its keyboard and controllers, **Pin to this player**, **Remap…** and **Swap** (*Swap with Player N*), plus **Other controllers** with **Use for Player N** buttons. While a Four Score game runs there are four cards, so you can put a controller on Player 3 or 4.

Remap… here works as in Settings (see Remapping buttons), with one more choice: **Only for** *the game's title* keeps the change for the running game alone. Closing the flyout stops Remap… waiting for a key.

Assist mode

Assist applies to the current session only and is always off when a game starts.

Option	Values
Assist mode	Turns the options below on.
Speed	Normal speed, Half speed, Quarter speed
Colors	Standard colors, Red-green (deuteranopia), Red-green (protanopia), Blue-yellow (tritanopia)
Unlimited rewind	Removes the rewind memory limit.
Reduce flicker	Draws every sprite on a line.

Practice mode

1. Open **Practice** and choose **Set practice point** where you want to start a run.
2. Press F6 (**Retry**) to go back to that point instantly.
3. Press F7 (**Split**) to mark the end of a segment.

The timer counts emulated frames, so it's exact even when the game slows down. GameVault keeps your personal best for each segment. Personal bests aren't recorded while codes are on.

Passwords

Many older games save progress with a password.

- **Automatic capture.** For games GameVault has a password definition for, passwords are saved automatically when the password screen appears, and a notice says *Password saved*. GameVault ships a definition for *Metroid* (USA); its password is saved each time it's shown after GAME OVER.
- **By hand.** Press `F9` on the password screen. The game pauses; type the **Password** and an optional **Note**, and choose **Save**.

Saved passwords are listed in **Game Details → Passwords** with a **Copy** button, and the newest one is shown on the *Previously on…* card.

Journal and maps

Each game has a journal in Game Details.

- **New note** creates a note with a rich-text editor.
- `F12` takes a screenshot and adds it to the journal. **Attach to note** links a picture to the selected note.
- **Add a map…** imports an image, for example a map you drew or scanned.
- For games with a memory map, GameVault records each room you visit and builds a map of where you've been automatically.

Game Genie and RAM codes

The **Codes** panel (NES games) is in the Player and in Game Details.

- Type a 6- or 8-letter **Game Genie** code, such as `SXIOPO`, or a **RAM code** such as `075A:09` (address:value, written every frame). The panel shows what it decodes to as you type. Add a description and choose **Add**.
- Switch codes on and off individually. `F3` turns them all off. Codes you've switched on are remembered for the game.
- **Import…** reads `.txt` lists (`CODE description` per line) and `.cht` cheat files from other emulators. **Export…** writes a `.txt` list.
- **Codes from GameGenie.com**: choose **Get codes from GameGenie.com** to download that game's codes from GameGenie.com (© Danworld, Inc.). Nothing is downloaded until you ask. This needs an internet connection.
- **Code collections** you imported in **Settings → Codes** show the codes they have for the game. If the game isn't matched, search for it in your collection; GameVault remembers your choice.

> **Note:** Practice personal bests aren't recorded while codes are on, codes are off in hardcore mode and during challenges, and netplay starts with codes off.

Challenges

A challenge is a file you share with a friend: a starting save state, a goal, and an optional time limit. The other player's attempt is recorded and can be replayed to prove the result. No server is involved.

Make a challenge (games with a memory map): open **Challenges**, give it a **Name**, pick a **Goal** (for example *Boss HP = 0*), optionally a **Time limit in seconds**, and choose **Save challenge...**. The challenge starts from this moment.

Take a challenge: choose **Open challenge...**, then **Start**. Rewind, loading states and codes are off until the attempt ends. When you finish, choose **Save result...** to send it back. **Check a result...** replays someone else's result file and confirms it.

Stat overlays

For games with a memory map, the **Stats** button draws values from the game, such as exact health or a boss's hit points, in the top-left corner. It's remembered per game and isn't included in screenshots.

Taking screenshots

F12 saves a screenshot to the game's journal.

Controls

GameVault needs no setup for controls. The keyboard always works as Player 1, and controllers are detected and mapped automatically when you plug them in, including Xbox, PlayStation, Switch Pro and 8BitDo pads.

The layouts below are the defaults. You can change any of them with **Remap…** (see Remapping buttons), choose which controller is which player, the turbo speed, and a two-player keyboard layout.

Keyboard

Player 1 (default)

Console button	Key
D-pad	Arrow keys
B	Z
A	X
Select	Right Shift
Start	Enter
Turbo B / Turbo A (NES)	A / S
C (Genesis)	C
X, Y, Z (Genesis)	Q, W, E
L, R (SNES; PlayStation and PlayStation 2 L1, R1; N64 L, R)	1, 2
L2, R2 (PlayStation and PlayStation 2); Z (N64)	3, 4
L3, R3 (PlayStation and PlayStation 2)	5, 6
Right stick (PlayStation and PlayStation 2; N64 C buttons)	I J K L
D-pad (PlayStation, PlayStation 2 and N64, where the arrows are the left stick)	T F G H

Each system's own layout is described in Supported systems.

Keyboard input reaches the game only while the game has focus. When you switch away from GameVault, every held key is released so nothing stays stuck down.

Two players, one keyboard

Player 2 has no keys by default. Turn on **Settings → Controllers → Two players, one keyboard** to split the keyboard:

Console button	Player 1 (right half)	Player 2 (left half)
D-pad	Arrow keys	W A S D
B	, (comma)	F
A	. (period)	G
Select	Right Shift	Q
Start	Enter	E

There are no turbo keys in this layout. Some keyboards can't register many keys at once and may drop presses when both players hold several buttons.

Controllers

Buttons are mapped by **position**, so the NES layout (B on the left, A on the right) is the same on every pad.

Console button	Pad position	Xbox label	PlayStation label
D-pad	D-pad and left stick		
B	Bottom	A	Cross
A	Right	B	Circle
Turbo B	Left	X	Square
Turbo A	Top	Y	Triangle
Select	Back	View	Share / Create
Start	Start	Menu	Options
Rewind (hold)	Left shoulder	LB	L1
Fast-forward (hold)	Right shoulder	RB	R1
Pause / overlay	Guide, or hold Back + Start for 1 second		

Systems with more buttons (Genesis, SNES, PlayStation, N64) switch to their own layout while their games run; see Supported systems.

Turbo buttons press and release the button repeatedly while held. Choose the rate in **Settings → Controllers → Turbo speed** (15 presses per second by default).

By default a pad can't press Left and Right (or Up and Down) at once, like a real NES pad; the most recent direction wins. **Settings → Advanced → Allow opposite directions** turns this off.

Which controller is which player

- The keyboard is always on Player 1.
- Controllers join in the order you connect them: the first joins the keyboard on Player 1, the second becomes Player 2. A short notice says, for example, *8BitDo SN30 Pro is now Player 2*.
- **Settings → Controllers** (and the Player's **Controllers** button) shows a card per player. **Swap players** swaps Players 1 and 2; each card's **Swap** menu swaps it with any other player. **Pin to this player** makes a controller always take that player when it connects. Extra controllers are listed under **Other controllers** with a **Use for Player N** button for each player.
- While a Four Score game runs there are four players: waiting controllers join as Players 3 and 4, and you can move any controller there with **Use for Player 3** / **Use for Player 4**.

If a controller disconnects

The game pauses and says which player's controller was lost. Reconnect it (it gets the same player back), or press A or Start on another controller to take over. Resume from the toolbar.

Using a controller in menus

Every connected controller can drive every page and menu, so you never need the mouse or keyboard to find and start a game. Buttons are named by position, so they're the same on Xbox, PlayStation and Switch-style controllers:

Button	In menus
D-pad or left stick	Move to the nearest item in that direction. Hold to keep moving. On a slider, left and right change its value.
Bottom (A / Cross)	Press the button, flip the switch or check box, open the list or section, or start the game
Right (B / Circle)	Close the open menu or list; otherwise go back a page
Top (Y / Triangle)	The item's menu, for example **Play** on a game tile
Left and right shoulders	Switch between Library and Settings
Left and right triggers	Jump about a screen up or down a long page. Hold to keep going.
Start	Go to the menu on the left

On a text box, such as the Library's search, the bottom button opens Windows' touch keyboard where Windows has one. On other PCs you still need a keyboard to type. While a game is playing, controllers go to the game. The **Guide** button, or **Back + Start** held for a second, brings up the toolbar, and controllers drive the menus again until you resume.

Remapping buttons

Every card in **Settings → Controllers** lists its keyboard and controllers, each with **Remap…**. It opens an editor for that keyboard or controller on that player:

1. Pick the **System**. The rows are that system's own buttons by their own names: *Button 1*, *Button 2* and *Pause* on the Master System; *A*, *B*, *C*, *X*, *Y*, *Z* and *Mode* on the Genesis; *Cross*, *Circle*, *Square*, *Triangle*, *L1*–*R2* on the PlayStation; *A*, *B*, *Z* and *C-up* to *C-right* on the N64, and so on.
2. Choose **Change** on a row and press the key, or the button on that controller. On the keyboard, Esc cancels.
3. The new key or button replaces the old one. A key or button does one thing, so if it was already used, that other row is left unassigned: *X was A; that's unassigned now.* **Clear** unassigns a row.

Changed rows are shown in bold. **Back to the standard layout** undoes every change for that system. Changes save and apply immediately, also in a game that's running.

- **Per player and per controller model.** A remap belongs to the keyboard, or to a model of controller (every *Xbox Wireless Controller* shares it), on one player. Other models keep their own layouts.
- **Per game.** In the Player, the **Controllers** button has the same editor with **Only for** *the game*: changes then apply to that game alone, on top of the layout for every game.
- **Keyboard layouts.** For the keyboard, **Layout** offers ready-made layouts for every system; choose **Use this layout**:

Button	Standard	One-handed, left hand	One-handed, right hand
D-pad (the stick on N64 and PS2)	Arrow keys	W A S D	I J K L
B / A	Z / X	Q / E	U / O
Start / Select	Enter / Right Shift	R / F	P / ;
Turbo B / Turbo A	A / S	1 / 3	7 / 9

Buttons a layout doesn't list keep their keys, unless the layout takes their key; then they're unassigned and you can give them one with **Change**.

GameVault's own keys can't be used: Esc, Backspace, Tab, F2–F9, F11 and F12 show *… is one of GameVault's own keys (save, load, rewind…). Choose another.* The controller's Guide button pauses and can't be remapped. On a controller the sticks and triggers stay as they are (the N64's Z on the triggers and C-up/C-right on the right stick are shown but fixed); every other button can move. A shoulder button that you give a game button no longer rewinds or fast-forwards.

Mouse

NES Zapper games use the mouse: aim with the pointer and click to fire. The Arkanoid paddle also uses the mouse: moving the pointer across the picture turns the knob, from the left edge to the right edge, and the left button fires. GameVault sets these up automatically for games its database knows; otherwise set **Game Details → Properties → Controls** to *Zapper* or *Arkanoid paddle (mouse)*.

Power Pad

Power Pad (Family Trainer) games are played on the keyboard, in the mat's rows of four. GameVault sets this up for the Power Pad games it knows; otherwise choose **Game Details → Properties → Controls → Power Pad (keyboard)**.

Mat	Keys
Buttons 1–4	U I O P
Buttons 5–8	J K L ;
Buttons 9–12	M , . /

None of these is a controller key, so Player 1's controller (or the arrows, Z and X) still works for the menus.

Hotkeys

These keys never reach the game.

Action	Keyboard	Controller
Pause / resume (in full screen, first leaves full screen)	Esc	Guide, or hold Back + Start
Full screen on/off	F11 or Alt + Enter	
Rewind (hold)	Backspace (Shift + Backspace on Apple II)	Left shoulder
Fast-forward (hold)	Tab	Right shoulder

Action	Keyboard	Controller
Back to before the last life lost	F2	
All codes off	F3	
Next disk side / next disc	F4	
Quick save / quick load	F5 / F8	
Practice: retry / split	F6 / F7	
Write down a password	F9	
Screenshot	F12	
Apple II Reset	Ctrl + F12	

Controller hotkeys work only from controllers assigned to a player.

Access keys

Press Alt to show letters on buttons, then press the letter.

Screen	Keys
Anywhere	**L** Library
Library	**A** Add a ROM folder, **O** Open game file, **I** Import catalog, **D** Dump cartridge, **S** Search, **V** List/grid view
Game Details	**P** Play
Player	**B** Back, **P** Pause, **R** Reset, **S** Save states, **A** Assist, **D** Disks, **C** Codes, **N** Netplay, **F** Full screen, **T** Stats, **H** Challenges, **E** Achievements, **O** Controllers

Settings reference

Open **Settings** with the gear at the bottom of the navigation pane. Changes save immediately; there's no Apply button. Some settings say *takes effect next time a game starts*.

If the current family profile isn't allowed to change settings, the page shows *Settings are locked for this profile* and a parent can unlock it with the PIN (see Family profiles).

Family

Profiles, the parent PIN and daily play limits. See Family profiles.

RetroAchievements

Setting	Values	Default
Username, **Password**, **Sign in**	Your RetroAchievements account	Signed out
Hardcore mode	On / Off	**On**

See RetroAchievements.

Library

Game folders lists the folders GameVault watches, each with **Remove**. **Add a folder…** adds one. **Look for new games** rescans them all. GameVault looks for games for every system it plays, including games inside `.zip` files. **Remove** asks first (*Remove this folder?*): the folder's games leave your library, but the files aren't deleted, and adding the folder again brings them back with their saves and notes.

Video

Setting	Values	Default
Hide screen edges	Show the whole picture, 4 lines, 8 lines (like most TVs), 12 lines, 16 lines, at the top and at the bottom. Hide more if a game draws flickering lines or garbage at the edges.	8 lines

Setting	Values	Default
Picture style	Sharp pixels, Smooth, Living room CRT, Arcade monitor	Sharp pixels
TV pixel shape	On / Off. Stretches the picture to the 8:7 pixel shape of a TV.	Off
Play in full screen	On / Off. Games start in full screen.	Off

Audio

Setting	Values	Default
Game volume	0 to 100	80
Mono	On / Off. Plays both stereo channels on both speakers, for one speaker or one ear.	Off

Sound is in stereo where the console had it: the Genesis / Mega Drive, Super Nintendo, PlayStation, Nintendo 64 and PlayStation 2. The NES, SG-1000, Master System and Apple II were mono, so their sound is the same on both speakers.

Gameplay

Setting	Values	Default
Reduce input lag	Off, Low, Medium, High (run-ahead of 0 to 3 frames)	Low
Pause when GameVault is in the background	On / Off	On
Rewind memory	64 MB, 128 MB, 256 MB, 512 MB, 1 GB	256 MB

Setting	Values	Default
Keep game saves next to the game files	On / Off. Cartridge saves go beside each game as a `.sav` with the same name (for example `Game.sav` next to `Game.nes`), so other emulators can use them. Saves already in GameVault's Saves folder are still found. If a game's folder can't be written to, its save stays in GameVault's folder. Memory cards and Controller Paks always stay there.	Off

Reduce input lag uses run-ahead: GameVault quietly runs a frame or more ahead so the game responds sooner. Games known to break with it have it turned off automatically, and you can override it per game in Game Details → Properties. Higher settings need a faster PC.

Controllers

Setting	Values	Default
Swap players	Swaps Player 1 and Player 2. Each card's **Swap** menu swaps it with any other player.	
Pin to this player / **Unpin**	Per controller	Not pinned
Remap...	Per keyboard or controller model, per player and per system. See Remapping buttons.	Standard layouts
Use for Player N	For extra controllers, one button per player (four during Four Score games)	
Two players, one keyboard	On / Off	Off
Turbo speed	30, 20, 15, 12, 10 or 7.5 presses per second	15

Cartridge dumper

The path to `inlretro.exe`, the INL Retro-Prog command-line tool, for example `C:\INL-retro\inlretro.exe`. Type or paste it, or choose **Choose...** to pick the file. Takes effect the next time GameVault starts. See Cartridge dumper.

Codes

Code collections you've imported, each with **Remove** (which asks first). **Import a code collection...** adds a `.xlsx` or `.csv` file. See Code collections.

Disk System

The path to your Famicom Disk System BIOS, for example `C:\NES\disksys.rom`, or **Choose...** to pick it.

Apple II

Setting	Example
System ROM	`C:\Apple\apple2e-enhanced.rom`
Disk II boot ROM	`C:\Apple\disk2.rom`

Each box has **Choose...** to pick the file.

Super Nintendo

Setting	Values	Default
Enhancement chip ROMs	A folder holding your own `dsp1b.rom` (or `dsp1.rom`), `dsp2.rom`, `dsp3.rom` or `dsp4.rom`, or **Choose...**	Empty

PlayStation

Setting	Values	Default
BIOS	Path, for example `C:\PlayStation\scph1001.bin`, or **Choose...**	Empty
Skip the startup screens	On / Off	Off
Analog mode	On starts the controllers in analog mode, as if you'd pressed the DualShock's Analog button. Takes effect the next time a game starts.	Off

PlayStation 2

Setting	Values	Default
BIOS	Path, for example `C:\PlayStation 2\scph39001.bin`, or **Choose...**. GameVault checks it's a 4 MB BIOS and shows its version.	Empty
Skip the startup screens	On / Off	Off

Nintendo 64

Setting	Values	Default
Expansion Pak	On / Off. Takes effect the next time a game starts.	On
Rumble Pak	On puts a Rumble Pak in every controller instead of the Controller Pak, so games can shake your controller. Takes effect the next time a game starts.	Off

Advanced

Setting	Values	Default
Allow opposite directions	On / Off. Lets the D-pad press Left+Right or Up+Down together.	Off

Restore defaults

Restore defaults resets Video, Audio, Gameplay and *Allow opposite directions*. It doesn't touch your folders, BIOS and ROM paths, the dumper path, controller settings, PlayStation and Nintendo 64 options, family profiles or RetroAchievements.

All path boxes save when you leave the box, or as soon as you pick a file with **Choose...**, which also works with a controller. Quotes around a pasted path are removed for you.

Online and multiplayer

GameVault works fully offline. These features use the network only when you ask them to.

Netplay

Two people on the **same local network** (home Wi-Fi or wired network) can play together, each on their own PC.

What you need:

- Both PCs on the same network. Internet play isn't available yet.
- **The same game file** on both PCs. GameVault checks this and never sends game data over the network.
- For Apple II games, the same Apple II ROM files on both PCs.
- RetroAchievements **Hardcore mode** off on both PCs.
- A game for any system except the PlayStation 2, whose console is too large for netplay's rollback.

Hosting:

1. Start the game.
2. Open **Netplay** and choose **Host a game on this network**.
3. Read the invite code to the other player, or choose **Copy code** and send it. It looks like `ABCDE-12345`.

Joining:

1. Start the same game.
2. Open **Netplay**, type the code under *Or join with the other player's code*, and choose **Join**.

Both games restart from the title screen, without battery saves or codes. The Netplay panel shows whether you're Player 1 or Player 2. Progress made during netplay isn't written to the game's save file. Choose **Leave netplay** to stop.

Firewall: the first time, Windows may ask whether GameVault can use the network. Allow it on **Private networks**. If you clicked *Cancel*, allow GameVault in *Windows Security → Firewall & network protection → Allow an app through firewall*. Netplay uses UDP port 47800, or another free port if that one is busy. Your network must be set to *Private* in Windows network settings; public networks (cafés, hotels, many university networks) often block devices from reaching each other.

Messages:

Message	What to do
That invite code isn't right.	Codes have 10 letters and digits. Case, spaces and dashes don't matter.
The other player has a different game or version.	Both players need the identical file. Compare the files or copy yours to them.
Lost the connection to the other player.	Check both PCs are on the same network and the firewall allows GameVault.
The two games stopped matching, so netplay stopped.	Start netplay again.
This PC doesn't seem to be on a network.	Connect to the network first.
Netplay is off in RetroAchievements hardcore mode.	Turn Hardcore off in Settings → RetroAchievements.

Famicom Disk System games can't be played over netplay yet.

RetroAchievements

RetroAchievements is a free community service that adds achievements and leaderboards to classic games.

Setup:

1. Create a free account at https://retroachievements.org.
2. In GameVault, open **Settings → RetroAchievements**, type your **Username** and **Password**, and choose **Sign in**. It shows *Signed in as …* with your points.

Your password is sent only to RetroAchievements. GameVault keeps an encrypted sign-in token instead, readable only by your Windows account. Each family profile signs in separately.

Playing: when a game RetroAchievements recognises starts, the **Achievements** button appears in the Player. Unlocks pop up in the bottom-right corner, and leaderboard attempts are announced as they start and finish.

Hardcore mode (on by default) follows RetroAchievements' rules: achievements count double, and loading save states, rewind, slow motion, codes and netplay are off. Saving states still works. Turning it on restarts the running game. Turn it off in Settings if you'd rather keep those features.

> **Note:** RetroAchievements hasn't yet approved GameVault for hardcore unlocks, so they may be recorded as softcore or refused until it does. If your copy says *The RetroAchievements component is missing*, it was built without CMake; see Installing GameVault.

Message	What to do
RetroAchievements didn't accept that username and password.	Check them on the RetroAchievements website.
Your RetroAchievements sign-in has expired.	Sign in again.
RetroAchievements didn't answer.	Check your internet connection.

GameGenie.com codes

In the **Codes** panel of an NES game, **Get codes from GameGenie.com** downloads that game's codes from GameGenie.com (© Danworld, Inc.) and shows them with a link to the page. Nothing is downloaded until you choose it. Pages are kept for up to 30 days so they work offline afterwards.

Code collections

If you have your own list of Game Genie codes in a spreadsheet, import it in **Settings → Codes → Import a code collection...**. It's copied into GameVault's data folder.

The file must be `.xlsx` or `.csv` with a header row containing these columns (other columns are ignored):

Column	Header text GameVault looks for
Game	`Game`
Code	`Code`, `Codes` or `Key in`
What it does	`Effect`, `Effects` or `Description`
Source (optional)	`Source` or `Code Source`

In a spreadsheet, GameVault reads the sheet named **NES**, or the first sheet. Rows whose code isn't valid are kept as notes for that game. A collection's codes then appear in the Codes panel for matching games.

Cartridge dumper (INL Retro-Prog)

GameVault can dump NES cartridges with an **INL Retro-Prog**, an open-source USB cartridge reader. This is optional.

Setup:

1. Install the INL Retro-Prog software and its **USB driver** from the INL website, following its instructions. Note the folder that contains `inlretro.exe`; GameVault runs it from that folder and uses its `scripts\inlretro2.lua`.
2. Plug in the INL Retro-Prog.
3. In **Settings → Cartridge dumper**, choose **Choose…** and pick `inlretro.exe`, or paste its full path, for example `C:\INL-retro\inlretro.exe`.
4. **Restart GameVault.**

Then use **Library → Dump cartridge**; see Dumping a cartridge. Supported boards: NROM, MMC1, UxROM, CNROM, MMC3, AxROM, MMC2, MMC4, Color Dreams, BNROM, GxROM and Sunsoft FME-7.

Message	What to do
GameVault can't find the INL Retro-Prog software.	Check the path in Settings and restart GameVault.
GameVault can't find the INL Retro-Prog. Check that it's plugged in…	Reconnect the USB cable.
Windows didn't let GameVault use the dumper.	Install the INL Retro-Prog USB driver.
The reads didn't match. Clean the cartridge contacts and try again.	Clean the contacts with isopropyl alcohol and reseat the cartridge.
The INL Retro-Prog can't read this kind of cartridge yet.	That board isn't supported.

Family profiles

Each person in your household can have their own profile, with their own library status, save states, battery saves, timeline, journal, codes and settings.

Setting up profiles

In **Settings → Family**:

1. Type a name under **New profile**.
2. Optionally set **Minutes a day** (5 to 1440) to limit daily play. Leave it empty for no limit.
3. Tick **Can change settings** if this person may change GameVault's settings.
4. Choose **Add**.

The first profile, *Player 1*, is the main profile and can't be removed. Neither can the profile that's in use.

Parent PIN

Type a PIN of 4 to 12 digits and choose **Set parent PIN**. Once a PIN is set:

- Profiles without *Can change settings* see Settings locked. A parent types the PIN and chooses **Unlock**.
- **Lock** locks Settings again. **Remove PIN** removes it.

> **Forgot the PIN?** GameVault stores only a scrambled (hashed) copy, so it can't show it to you. To clear it, close GameVault, open `%LOCALAPPDATA%\GameVault\profiles.json` in Notepad, delete the `"pin": { … }` entry (including its trailing comma if one follows), and save. Profiles and saves are kept.

Switching profiles

When there's more than one profile, a profile button appears at the bottom of the navigation pane. Choose it, then choose a name under **Who's playing?**. Switching closes the running game (it's saved first) and reopens GameVault with that person's games and saves.

Daily play limits

For a profile with a daily limit:

- A notice appears 5 minutes before the end: *5 minutes of play left today.*
- At the limit, the game is saved and paused, and a *Time's up for today* card appears.
- A parent can type the **Parent PIN** and choose **Keep playing**. That profile then isn't stopped again for the rest of the day.

Each extra profile keeps its files in `%LOCALAPPDATA%\GameVault\Profiles\<profile id>\`; the main profile uses `%LOCALAPPDATA%\GameVault\` itself.

Your data

Where GameVault keeps things

Everything GameVault creates is in `%LOCALAPPDATA%\GameVault\` (usually `C:\Users\<you>\AppData\Local\GameVault\`). Your ROM folders are only ever read, never written to (unless you turn on **Keep game saves next to the game files**, which writes each cartridge save beside its game), and your BIOS files are read from where you put them.

Path	Contents
`settings.json`	Your settings, including your button remaps
`library.db`	Library: games found, catalog, ownership, status, ratings, notes, playtime
`Saves\`	Battery saves (`.sav`), Disk System and Apple II disk writes (`.fdssave`, `.a2save`), PlayStation memory cards (`.mcd`), N64 Controller Paks (`.mpk`), PlayStation 2 memory cards (`.ps2`)
`States\`	Save states and their thumbnails
`Passwords\`	Captured passwords
`Journal\`	Notes, screenshots and maps
`Cheats\`	Your codes; `Collections\` holds imported code collections, `GameGenie.com\` the downloaded pages
`Sessions\`, `Maps\`, `Practice\`	Recent play sessions, automatic maps, practice personal bests
`retroachievements.json`	Encrypted RetroAchievements sign-in
`profiles.json`	Family profiles and the parent PIN (hashed)
`Profiles\<id>\`	The same files for each extra family profile
`bios\`	Copies of the BIOS files you chose with **Choose...** (`fds\`, `ps1\`, `ps2\`), shared by every family profile
`logs\crash.log`	Details of any crash

Cover pictures stay where you chose them; only their location is stored, so don’t move or delete them.

Backing up and moving to a new PC

1. Close GameVault.
2. Copy the whole `%LOCALAPPDATA%\GameVault\` folder to a backup drive.
3. On the new PC, install GameVault, copy the folder back to the same place, and copy your games and BIOS files.
4. If your games or BIOS files are in different places on the new PC, update **Settings → Library** and the BIOS paths. Saves are matched to games by contents, not by file name or location, so they carry over.

The RetroAchievements sign-in is tied to your Windows account, so sign in again on the new PC.

Resetting GameVault

- **Settings only: Settings → Restore defaults**, or close GameVault and delete `settings.json`.
- **Everything:** close GameVault and delete (or rename) the `%LOCALAPPDATA%\GameVault\` folder. This removes your saves too, so back it up first.

Troubleshooting

Problem	Try this
A game isn't in the library	Check its extension is listed in Supported systems. A `.zip` must hold exactly one game; files over 64 MB and PlayStation `.bin` files without a `.cue` are skipped. Choose **Settings → Library → Look for new games**.
That file isn't a game GameVault can open.	The file is damaged, has the wrong extension, or is for a system GameVault doesn't support.
GameVault couldn't open that file.	It was moved, deleted, or is open in another program.
A game shows a BIOS or ROM message	Set the file in Settings for that system. See What you must supply.
A game has the wrong name or shows as Unknown	Import a catalog for that system, or type the title and details in Game Details → **Edit details**. Bad or modified dumps don't match the catalog.
A key or button does the wrong thing	It may be remapped. Open **Remap…** for that keyboard or controller and choose **Back to the standard layout**; in the Player, check whether a remap is set **Only for** this game.
A European game runs too fast, or a US game too slow	Set **Game Details → Properties → Region**.
An Apple II disk shows only the startup screen	Set the Disk II boot ROM in **Settings → Apple II**.
No sound	Check **Settings → Audio → Game volume** and the Windows volume mixer. Sound is muted while fast-forwarding.
Sound or picture stutters	Close other heavy programs, set **Reduce input lag** to *Low* or *Off*, and choose **Sharp pixels**. The heaviest Nintendo 64 3D scenes can run below full speed on a CPU with few cores.
The game pauses when I click another window	That's **Pause when GameVault is in the background**; turn it off in Settings → Gameplay.

Problem	Try this
Controller doesn't work	Make sure GameVault is the active window; controllers are ignored in the background. Check **Settings → Controllers** for which player it's assigned to. If it's under *Other controllers*, choose **Use for Player 1**. Some apps (Steam, Game Bar) capture the Guide button; hold **Back + Start** instead.
A keyboard key does nothing	Click the game picture to give it focus. Two-player keyboard games: some keyboards can't register many keys at once.
Rewind or save states are greyed out	RetroAchievements **Hardcore mode** or a challenge attempt turns them off.
A game freezes or glitches with Reduce input lag on	Set **Game Details → Properties → Reduce input lag** to *Off* for that game.
NES sprites flicker	Turn on **Game Details → Properties → Reduce flicker**, or Assist → **Reduce flicker**.
An N64 game won't start	Turn off **Settings → Nintendo 64 → Expansion Pak**.
An N64 game doesn't keep its save	Its save type may be missing from GameVault's database. Try the cartridge's save type in **Game Details → Properties → Save type**.
A PlayStation game won't start with fast boot	Turn off **Skip the startup screens**.
A PlayStation 2 game won't start with fast boot	Turn off **Settings → PlayStation 2 → Skip the startup screens**.
A PlayStation 2 game runs slowly	Many games still run below full speed (see PlayStation 2, *Performance*). If you built GameVault yourself, use a Release build (see Installing GameVault). Close other heavy programs.
A PlayStation 2 game isn't in the library	Discs are recognised by their SYSTEM.CNF file; check the image isn't damaged. A `.bin` must come with its `.cue`.
Settings are greyed out	The family profile is locked. A parent can unlock it with the PIN.
GameVault crashed	Details are in `%LOCALAPPDATA%\GameVault\logs\crash.log`. Include that file when you report the problem.

Command-line options

Option	What it does
`--sample-data`	Starts with a made-up library so you can explore without games. Nothing is saved.
`--sample-data=empty`	The same, with an empty library, to see the first-run screens.
`--page=<name>`	Opens on `Library`, `Settings`, `Player`, `GameDetails`, `CatalogImport` or `DumpCartridge`. Mainly for previews with `--sample-data`.
`--profile=<id>`	Opens as a family profile. GameVault uses this itself when you switch profiles.

GameVault can’t be given a game file on the command line yet.

Edition history

Edition	Date	Changes
1	October 2026	First edition. Covers NES and Famicom Disk System, SG-1000 and Master System, Apple II, Genesis / Mega Drive, Super Nintendo, PlayStation and Nintendo 64.
2	October 2026	Adds the PlayStation 2: its section in Supported systems, the DualShock 2's controls (both sticks, L3 and R3), Settings → PlayStation 2, `.ps2` memory cards, and its limits on rewind, run-ahead and netplay.
3	October 2026	Remapping buttons (per system, player, controller model and game; one-handed keyboard layouts), the Player's Controllers flyout and Players 3–4, the Power Pad and Arkanoid paddle, the bundled homebrew games, accessory profiles, memory maps and Metroid's password definition, per-system Game Details properties and details for unknown games, confirmations before deleting, Choose… buttons for BIOS files, saves next to the game files, the Hide screen edges amount, the rewind scrub bar, and the Disk II boot ROM message.
4	October 2026	A BIOS chosen with **Choose…** is copied into GameVault's own `bios` folder. The PlayStation 2 runs much faster (its main processor's code is translated as it runs, and its graphics are drawn on several cores), and the Nintendo 64's graphics are drawn on several cores too.

Edition	Date	Changes
5	October 2026	Every page and menu can be used with a controller alone: the D-pad moves between items, the bottom button presses, flips or opens them, the top button opens an item’s menu, the shoulders switch between Library and Settings, and the triggers jump a screen up or down a long page.
6	October 2026	Sound is in stereo on the Genesis / Mega Drive, Super Nintendo, PlayStation, Nintendo 64 and PlayStation 2, and **Settings → Audio → Mono** plays both channels on both speakers.
7	October 2026	Super Nintendo games with a Super FX chip (Star Fox, Yoshi’s Island) play, and games with a DSP chip (Super Mario Kart, Pilotwings) play with your own chip ROM in **Settings → Super Nintendo → Enhancement chip ROMs**.
12	October 2026	Nintendo 64: **Settings → Nintendo 64 → Rumble Pak** puts a Rumble Pak in every controller, and games shake your controller through it; games in 32-bit colour show all their colours.
14	October 2026	PlayStation: the controllers are DualShocks, with both sticks, L3 and R3, vibration and **Settings → PlayStation → Analog mode**; the GunCon and the mouse (chosen per game in Game Details); 24-bit videos show all their colours.